System Model Comparison

Two-stage obstacle detection pipeline — realistic timings per format and precision level

Hardware: NVIDIA RTX 1070 (8 GB VRAM) · Values marked 'Measured' are real benchmark results; others are estimates scaled proportionally.

Stage (Role)	Model	Format / Precision	Params	Est. RAM Usage	Processing Time (TTFT)	Est. FPS	GGUF?	Core Capability (What does it do?)	CoT (Reason?)
STAGE 1 — Eye / Filter | Florence-2-base | WITH GGUF (Quantized)									
Stage 1 Eye / Filter	**Florence-2-base**	**GGUF Q4_K_M** 4-bit quant.	~0.23B	~130 MB	**~0.8 – 1.2 sec** Measured · RTX 1070 (Optimized)	~0.9 FPS	**YES**	Detects obstacles, returns Bounding Box coordinates. Task-focused, no reasoning.	**No.** Pre-defined labels only (e.g. <OD>).
Stage 1 Eye / Filter	**Florence-2-base**	**GGUF Q8_0** 8-bit quant.	~0.23B	~230 MB	**~1.0 – 1.5 sec** Est. · RTX 1070	~0.75 FPS	**YES**	Same detection as Q4 with marginally higher accuracy. Slightly more VRAM.	**No.** Pre-defined labels only.
STAGE 1 — Eye / Filter | Florence-2-base | WITHOUT GGUF (FP32 Full Precision)									
Stage 1 Eye / Filter	**Florence-2-base**	**Non-GGUF FP32** Full precision	~0.23B	~900 MB	**~2.5 – 4.0 sec** Est. · RTX 1070	~0.3 FPS	**NO**	Same capability — full precision weights. Much heavier on VRAM, slower on edge hardware.	**No.** Same label-only behavior, higher resource cost.
STAGE 2 — Brain / Router | Qwen 3.5-VL | WITH GGUF (Quantized)									
Stage 2 Brain / Router	**Qwen 3.5-VL**	**GGUF Q4_K_M** 4-bit quant.	~0.8B	~700 – 900 MB	**~3.0 sec** Est. · RTX 1070	~0.33 FPS	**YES**	Analyzes Florence's coordinates + image. Plans escape route with step-by-step reasoning.	**Yes.** CoT: *"Turn right to avoid obstacle at X,Y."*
Stage 2 Brain / Router	**Qwen 3.5-VL**	**GGUF Q8_0** 8-bit quant.	~0.8B	~1.1 – 1.3 GB	**~4.5 – 5.5 sec** Est. · RTX 1070	~0.2 FPS	**YES**	Same routing logic, better output quality than Q4. Approaching VRAM limit on RTX 1070.	**Yes.** Full CoT, slightly better accuracy than Q4.
STAGE 2 — Brain / Router | Qwen 3.5-VL | WITHOUT GGUF (FP16) · Custom Self-Optimized vs. Default									
Stage 2 Brain / Router	**Qwen 3.5-VL**	**Non-GGUF FP16** Custom Optimized	~0.8B	~2.0 – 2.5 GB	**~3.0 sec** Measured · RTX 1070 Self-optimized (flash-attn, bf16 cast, custom loader)	~0.33 FPS	**NO**	**Full precision** — same speed as GGUF Q4 thanks to custom optimization. Superior output quality vs any GGUF variant. No quantization loss.	**Yes.** Full CoT at full precision. Best accuracy in the pipeline.
Stage 2 Brain / Router	**Qwen 3.5-VL**	**Non-GGUF FP16** Default (No Optimization)	~0.8B	~2.0 – 2.5 GB	**~10 – 14 sec** Measured · RTX 1070 (Non-opt.) 3–4x slower than custom-optimized version	~0.08 FPS	**NO**	**Default FP16 load** — no flash-attn, no bf16 casting, no custom loader. Same weights as optimized version but 3–4× slower. Unusable for real-time.	**Yes.** Full CoT — but latency makes it completely unsuitable for real-time use.

GGUF Q4_K_M: 4-bit quantization — smallest footprint, fastest inference, minimal accuracy loss. Best for real-time edge use. **GGUF Q8_0**: 8-bit quantization — better accuracy than Q4, ~1.5–2× more RAM, moderately slower. **Non-GGUF FP16 Optimized**: full precision with flash-attention / memory optimizations enabled. **Non-GGUF FP16 Non-optimized**: default FP16 load — no flash-attn, significant latency penalty. TTFT = Time To First Token.